

1.2 Motion - Mark scheme

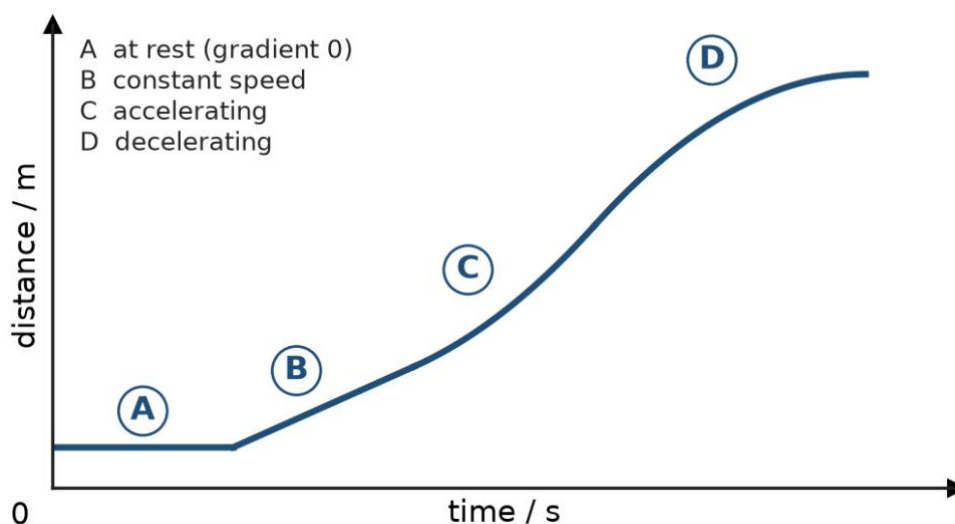
Cambridge IGCSE Physics (0625) | Syllabus 2026–2028

Tier: Core and Extended

For teachers. Contains the answers to Sections 6 and 7 and model versions of the Section 4 diagrams.

Model diagrams (Section 4)

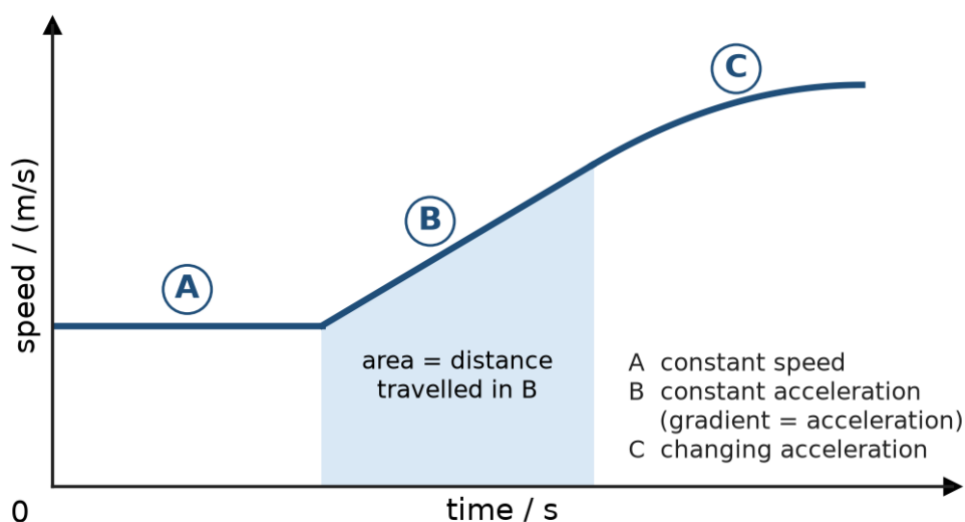
4.1 Distance–time graph **Core**



Gradient of a distance–time graph = speed. The line never slopes downwards for a journey in one direction.

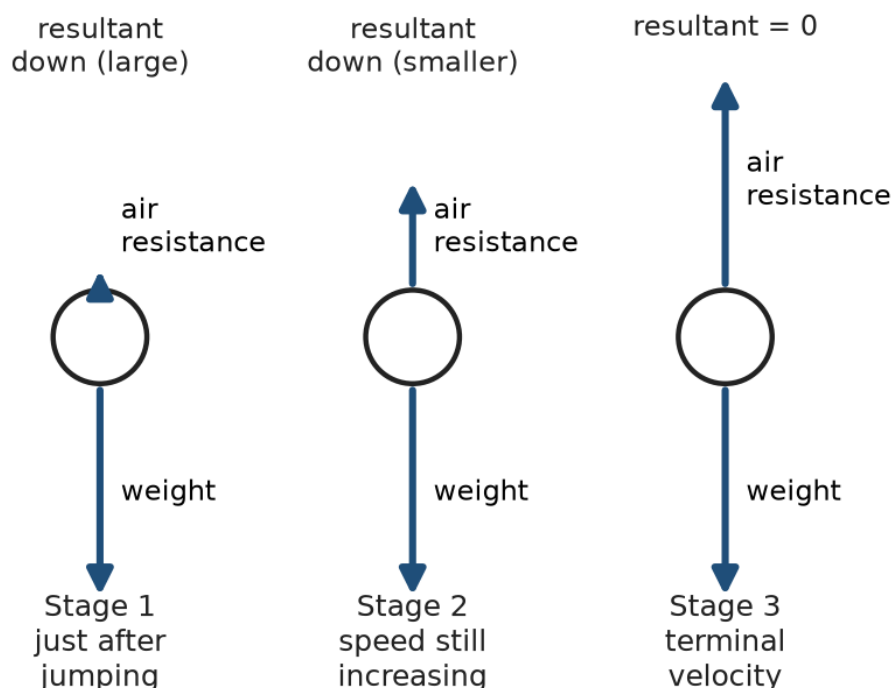
Credit for: A is a horizontal line (gradient zero); B is a straight line sloping upwards; C curves upwards (gradient increasing); D curves over (gradient decreasing); Stages labelled A, B, C, D.

4.2 Speed–time graph **Core**



Gradient = acceleration; area under the line = distance travelled. Identifying changing acceleration (stage C) is Extended.

Credit for: A is a horizontal line above the time axis; B is a straight line sloping upwards; Extended: C is a curve (gradient changing); Area under B shaded and labelled 'distance'.

4.3 Forces on a falling object **Extended**

Weight is constant. Air resistance increases with speed, so the resultant (downward) force and the acceleration decrease until air resistance = weight at terminal velocity.

Credit for: Weight arrow pointing down, the same length at every stage; Air resistance arrow pointing up, getting longer from stage 1 to stage 3; Stage 3: air resistance arrow equal in length to weight; Both forces labelled.

Multiple-choice answers (Section 6)

Each wrong option is a specific mistake. Where an examiner report records candidates making it, the series and paper are given in brackets. Use the third column to diagnose, not just to mark.

#	Answer	Why each wrong option is wrong
1	B	A This is speed. It leaves out the direction. C This is acceleration. D Treats velocity as a change in speed. (June 2024, Paper 12)
2	B	A Uses the uphill stage only ($3000 \text{ m} \div 600 \text{ s}$). C Averages the two speeds, $(5.0 + 10) \div 2$, instead of total distance $\div$ total time. (June 2021, Paper 23) D Uses the downhill stage only ($3000 \text{ m} \div 300 \text{ s}$).
3	D	A Picks the graph that looks the same as the speed–time graph. A horizontal distance–time line means the car is at rest. (June 2023, Paper 12) B Gradient increasing: the car would be accelerating. C Distance from the start decreasing: the car would be coming back.
4	C	A Works out the gradient (the acceleration, $20 \div 8.0$) instead of the area. B Reads the final speed as if it were a distance. D Multiplies the final speed by the time (20×8.0): a rectangle, not the triangle under the line. (Nov 2022, Paper 13 and June 2023, Paper 12)

#	Answer	Why each wrong option is wrong
5	A	B Thinks the acceleration of free fall increases as the object falls. (March 2022, Paper 12 and Nov 2024, Paper 21) C Thinks the acceleration of free fall decreases as the object falls. (Nov 2024, Paper 11) D Terminal velocity needs air resistance, and the question says it is negligible.
6	C	A Divides the top speed by the whole time on the graph ($8.0 \div 9.0$). B Includes the first second, when the athlete is still at rest ($8.0 \div 5.0$). The acceleration only lasts from $t = 1.0$ s to $t = 5.0$ s. D Works out the area under the sloping line ($\frac{1}{2} \times 4.0 \times 8.0$) instead of the gradient. (June 2024, Paper 22)
7	D	A Multiplies the change in velocity by the time (10×0.20) instead of dividing. B Subtracts the speeds ($6.0 - 4.0 = 2.0$ m/s) and ignores the change of direction. The change in velocity is $4.0 - (-6.0) = 10$ m/s. (March 2021, Paper 22 and June 2021, Paper 21) C Divides the average of the two speeds by the time ($5.0 \div 0.20$).
8	A	B Thinks the speed keeps increasing after terminal velocity is reached. (June 2022, Paper 23) C True only just after a parachute opens, when the skydiver slows down. D Confuses a zero resultant force with no forces. Weight and air resistance both act and are equal.

Mark scheme (Section 7)

Question 1 [1]

$$g = 9.8 \text{ m/s}^2 \text{ [1]}$$

Not 10 m/s^2 . The question asks for an acceleration, so the unit is m/s^2 ; 9.8 N/kg is gravitational field strength (1.3 Mass and weight).

Question 2 [2]

Speed is distance travelled per unit time / has magnitude only (scalar) [1]

Velocity is speed in a given direction (vector) [1]

Question 3 [3]

Converts units: $150 \text{ km} = 150\,000 \text{ m}$ and $2.5 \text{ h} = 9000 \text{ s}$ [1]

average speed = $150\,000 / 9000$ [1]

= 16.7 m/s (accept 17 m/s) [1]

Question 4 [5]

(a) First section (rectangle): $12 \times 6.0 = 72 \text{ m}$ [1]

Second section (trapezium): $\frac{1}{2} \times (12 + 30) \times 6.0 = 126 \text{ m}$ [1]

Total distance = $72 + 126 = 198 \text{ m}$ [1]

(b) $a = \Delta v / \Delta t = (30 - 12) / (12 - 6.0) = 18 / 6.0$ [1]

$a = 3.0 \text{ m/s}^2$ [1]

Question 5 [2]

A curve whose gradient decreases (becomes less steep) [1]

The line does not have to become horizontal to gain this mark.

The gradient is the speed, so the speed is decreasing [1]

Question 6 [3] Extended

Acceleration is the change in velocity per unit time (or rate of change of velocity) [1]

Do not credit a definition that mixes 'rate of change' with 'per unit time'.

$$a = (22.0 - 4.0) / 3.0 = 18.0 / 3.0 \quad [1]$$

$$a = 6.0 \text{ m/s}^2 \quad [1]$$

Question 7 [8] Extended

(a) Speed increases at a constant rate / constant acceleration (of $g = 9.8 \text{ m/s}^2$) [1]

(b) As the speed increases, the air resistance increases [1]

The resultant (downward) force decreases [1]

So the acceleration decreases [1]

Terminal velocity when air resistance = weight / resultant force = 0 (so acceleration = 0) [1]

Weight stays constant: do not credit 'gravity decreases' or 'weight decreases'.

(c) Any three of the four points below, max 3.

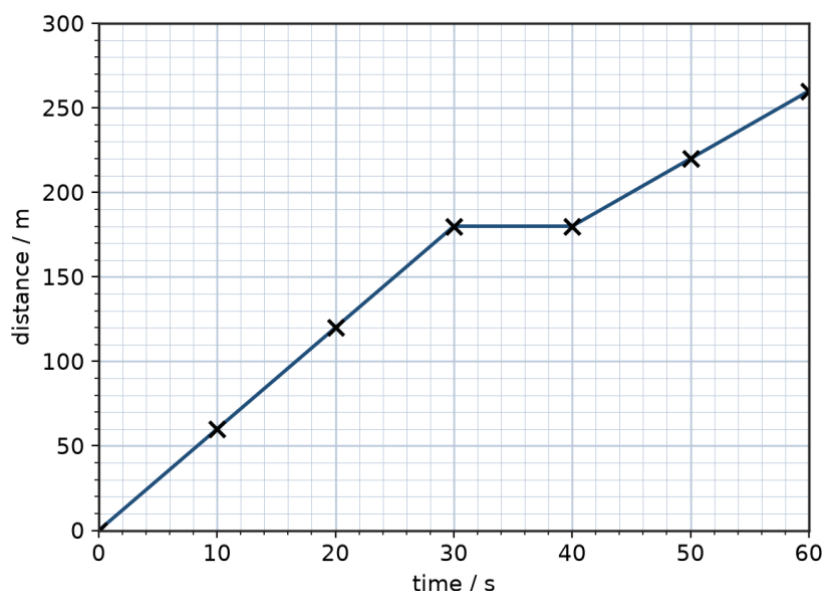
Air resistance increases (larger surface area) and becomes greater than the weight [1]

Resultant force is upwards, so the skydiver decelerates / slows down [1]

As the speed decreases, the air resistance decreases (until it equals the weight again) [1]

New terminal velocity, lower than before [1]

Do not credit 'the skydiver moves upwards': the velocity stays downwards throughout.

Question 8 [8]

(a) All seven points plotted to within half a small square [2]

One point wrong: 1 mark.

Straight lines joining the points with a ruler [1]

(b) Stationary / at rest [1]

(c) speed = gradient = $180 / 30$ [1]

$$= 6.0 \text{ m/s} \quad [1]$$

(d) average speed = $260 / 60$ [1]

$$= 4.3 \text{ m/s} \quad [1]$$

Question 9 [3] Extended

$$a = \Delta v / \Delta t = (0 - 8.0) / 2.5 \quad [1]$$

$$a = -3.2 \text{ m/s}^2 \quad [1]$$

Negative sign: the acceleration is opposite to the motion / the cyclist is decelerating (slowing down) [1]

Question 10 [6] Extended

(a) P [1]

(b) Speed increasing [1]

Acceleration decreasing / changing (gradient decreasing) [1]

(c) $a = \text{gradient} = 12 / 4.0$ [1]

$= 3.0 \text{ m/s}^2$ [1]

(d) 0 / zero (constant speed) [1]

Syllabus coverage

Where each checklist statement is practised in this booklet.

#	Statement	Practised in
1	Define speed as distance travelled per unit time; recall and use the equation $v = s/t$	WE1, Q3, Q8(c), MCQ2
2	Define velocity as speed in a given direction	Q2, MCQ1
3	Recall and use the equation: average speed = total distance travelled / total time taken	Q3, Q8(d), MCQ2
4	Sketch, plot and interpret distance–time and speed–time graphs	Diagrams 4.1, 4.2, Q8(a), Q10, MCQ3
5	Determine, qualitatively, from given data or the shape of a distance–time graph or speed–time graph when an object is: (a) at rest (b) moving with constant speed (c) accelerating (d) decelerating	Diagram 4.1, Q5, Q8(b), MCQ3
6	Calculate speed from the gradient of a straight-line section of a distance–time graph	WE1, Q8(c)
7	Calculate the area under a speed–time graph to determine the distance travelled for motion with constant speed or constant acceleration	WE2, Q4(a), MCQ4
8	State that the acceleration of free fall g for an object near to the surface of the Earth is approximately constant and is approximately 9.8 m/s^2	Q1, MCQ5
9	Define acceleration as change in velocity per unit time; recall and use the equation $a = \Delta v / \Delta t$	Q6, MCQ7
10	Determine from given data or the shape of a speed–time graph when an object is moving with: (a) constant acceleration (b) changing acceleration	Diagram 4.2, Q10(a), (b)
11	Calculate acceleration from the gradient of a speed–time graph	WE3, Q4(b), Q10(c), MCQ6
12	Know that a deceleration is a negative acceleration and use this in calculations	Q9, MCQ7
13	Describe the motion of objects falling in a uniform gravitational field with and without air/liquid resistance, including reference to terminal velocity	Diagram 4.3, Q7(a), (b), (c), MCQ8